MSG No. 95.142 P_I4_322 [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{7}{8}\right]$	`[1,8,14,15]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{5}{8}\right]$	`[2,4,11,13]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{8}\right]$	`[3,5,10,12]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{8}\right]$	`[6,7,9,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{8}\right]$	`[1,8,14,15]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{8}\right]$	`[2,4,11,13]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{5}{8}\right]$	`[3,5,10,12]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{7}{8}\right]$	`[6,7,9,16]`

Table 3: Wyckoff site: `8c`, site symmetry: `2'..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,14]`
2	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{3}{4}\right]$	`[2,11]`
3	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{4}\right]$	`[3,10]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[4,13]`
5	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[5,12]`
6	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[6,9]`
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[7,16]`
8	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}-z\right]$	`[8,15]`

Table 4: Wyckoff site: `8d`, site symmetry: `..2`

No.	position	mapping
1	$\left[x, -x, \frac{7}{8}\right]$	`[1,8]`
2	$\left[x, x, \frac{5}{8}\right]$	`[2,4]`
3	$\left[-x, -x, \frac{1}{8}\right]$	`[3,5]`
4	$\left[-x, x, \frac{3}{8}\right]$	`[6,7]`
5	$\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{3}{8}\right]$	`[9,16]`
6	$\left[x+\frac{1}{2}, x+\frac{1}{2}, \frac{1}{8}\right]$	`[10,12]`
7	$\left[\frac{1}{2}-x, \frac{1}{2}-x, \frac{5}{8}\right]$	`[11,13]`
8	$\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{7}{8}\right]$	`[14,15]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{7}{8}\right]$	`[1,15]`
2	$\left[\frac{1}{2}-x,\, x,\, \frac{5}{8}\right]$	`[2,13]`
3	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{8}\right]$	`[3,12]`
4	$\left[x,\, \frac{1}{2}-x,\, \frac{5}{8}\right]$	`[4,11]`
5	$\left[-x,\, x+\frac{1}{2},\, \frac{1}{8}\right]$	`[5,10]`
6	$\left[-x,\, \frac{1}{2}-x,\, \frac{3}{8}\right]$	`[6,16]`
7	$\left[x+\frac{1}{2},\, x,\, \frac{3}{8}\right]$	`[7,9]`
8	$\left[\frac{1}{2}-x,\, -x,\, \frac{7}{8}\right]$	`[8,14]`

Table 6: Wyckoff site: `8f`, site symmetry: `.2.`

No.	position	mapping
1	$[0,\, y,\, 0]$	`[1,5]`
2	$\left[-y,\, 0,\, \frac{3}{4}\right]$	`[2,8]`
3	$\left[y,\, 0,\, \frac{1}{4}\right]$	`[3,7]`
4	$\left[0,\, -y,\, \frac{1}{2}\right]$	`[4,6]`
5	$\left[\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	`[9,13]`
6	$\left[\frac{1}{2}-y,\, \frac{1}{2},\, \frac{1}{4}\right]$	`[10,16]`
7	$\left[y+\frac{1}{2},\, \frac{1}{2},\, \frac{3}{4}\right]$	`[11,15]`
8	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, 0\right]$	`[12,14]`

Table 7: Wyckoff site: `16g`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[-y,\, x,\, z+\frac{3}{4}\right]$	`[2]`
3	$\left[y,\, -x,\, z+\frac{1}{4}\right]$	`[3]`
4	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	`[4]`
5	$[-x,\, y,\, -z]$	`[5]`
6	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	`[6]`
7	$\left[y,\, x,\, \frac{1}{4}-z\right]$	`[7]`
8	$\left[-y,\, -x,\, \frac{3}{4}-z\right]$	`[8]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[9]`
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{4}\right]$	`[10]`
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{3}{4}\right]$	`[11]`
12	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	`[12]`
13	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[13]`
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	`[14]`
15	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{3}{4}-z\right]$	`[15]`
16	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{4}-z\right]$	`[16]`